# Experiment Six D3: Borneol to Camphor

Hannah Sommers

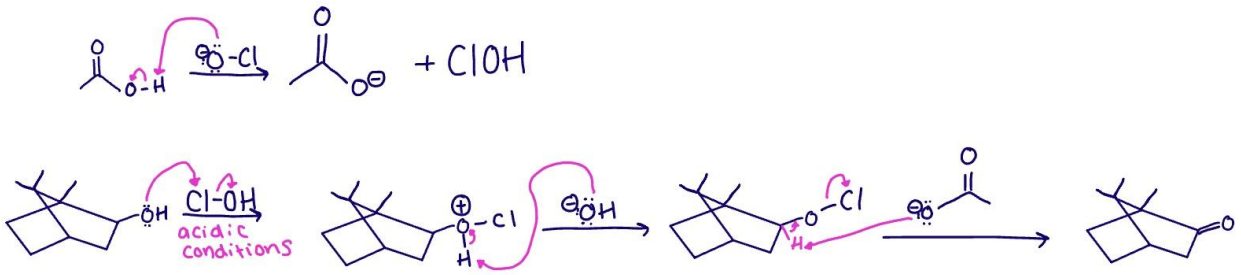
November 28, 2023

CHEM 2081-003

**Introduction**

In this experiment, the goal is to synthesize camphor from camphene. More specifically for the day of this experiment, borneol was transformed into camphor using a simple reflux in the presence of bleach.

The mechanism for Borneol to Camphor:




**Materials and Methods**

To start the experiment, a racemic mixture of borneol product and glacial acid was added to a long-neck round-bottom flask. The borneol was then dissolved in acetone with a boiling chip. A reflux apparatus was then set up in a water bath with temperatures between 40-50 C. Once all reagents were dissolved, aqueous sodium hypochlorite solution was added. The addition was done through the top opening of the condenser and paced slowly dropwise. After each addition of the hypochlorite, the mixture was given a gentle swirl to ensure all chemicals were mixed thoroughly. The reaction continued for 15 minutes after the last bleach addition. Once the mixture was cooled to room temperature, the camphor was extracted with two portions of methylene chloride. A centrifuge tube was used to make the extraction more efficient. The combined methylene chloride layers were saturated with sodium bicarbonate to remove excess bleach. The production of CO2 bubbles after every addition of the bicarbonate solution determined the presence of acidic residuals. The organic solution was washed sequentially with

5% sodium bisulfite and water. The organic solution was dried over anhydrous sodium sulfate. The solution was considered dry when freely rolling crystals of the drying agent were observed. The solution was evaporated with DCM over a water bath with a temp not exceeding 50 C. The collected solids were weighed and the percent yield was determined.

**Results**

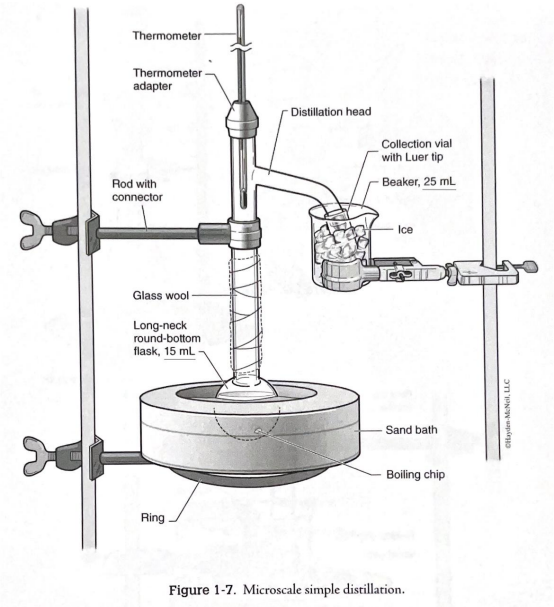



Figure 1-7. Microscale simple distillation.

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| | |
|---|---|
| **Weight of Dry Watch Glass (g)** | 32.737 |
| **Weight of Dry Watch Glass with Camphor (g)** | 32.986 |
| **Weight of Camphor (g)** | 0.249 |

Table 1. Weight of watch glass, Dry Watch Glass with Camphor, and Camphor.

The melting point was not calculated.

Percent yield: (Practical yield/ theoretical yield) X 100= (0.249/0.297) X 100= **83.84%**

Practical yield in grams: **0.249 of Camphor**

Theoretical yield in grams: **0.297 of Camphor**

**Calculations**

Molecular weight:

Borneol: $C_{10}H_{18}O$= (12.011 X 10) + (1.008 X 18) + (15.999 X 1)= **154.253 g/Mol**

Camphor: $C_{10}H_{16}O$= (12.011 X 10) + (1.008 X 16) + (15.999 X 1)= **152.237 g/Mol**

0.301 g of Borneol X ( 1 Mol / 154.253 g)= **0.00195134 Mol Borneol used to start**

0.00195134 Mol Borneol (152.237 g of Camphor/ 1 Mol of Camphor)= **0.297 g of Camphor**

**Discussion**

To start the experiment a racemic mixture of 0.301 grams of borneol product and 0.3 mL of glacial acid were combined in a long-neck round-bottom flask. Acetone and a boiling chip were added to the solution. A reflux apparatus was then set up and the water bath was kept at temperatures between 40-50 C. Once all reagents were dissolved, 6 mL of aqueous sodium hypochlorite solution was added. The addition was done through the top opening of the condenser and paced slowly dropwise. After each addition of the hypochlorite, the mixture was given a gentle swirl to ensure all chemicals were mixed thoroughly. The reaction continued for 15 minutes after the last bleach addition. Once the mixture was cooled to room temperature, the product was extracted with two portions of methylene chloride. A centrifuge tube was used to make the extraction more efficient. The combined methylene chloride layers were saturated with sodium bicarbonate which was then neutralized. The production of $CO_2$ bubbles after every addition of the bicarbonate solution determined the presence of acidic residuals. The organic solution was washed sequentially with 5% sodium bisulfite and water to remove any extra

bicarbonate solution. The organic solution was dried over anhydrous sodium sulfate to remove extraneous water. The solution was considered dry when freely rolling crystals of the drying agent were observed.  The DCM solution was evaporated over a water bath with a temperature not exceeding 50 C. The collected solids were weighed and the percent yield was determined.

**References**

Ruychev, Veselin. *CHEM 2081 ORGANIC CHEMISTRY II LAB MANUAL 2023 EDITION Department of Chemistry| Auburn University*. Department of Chemistry| Auburn University, 2023. (pg. 69-70)

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DCM density: 1.33 gr/mL

Organic Chemistry lab

24 October 2023

# Experiment Six Day 3 : Conversion of Camphene to Camphor

Safety:

* Never move chemical bottles out of the designated area. Never leave containers and/or bottles open after using them.

We will be working with camphene, sulfuric acid, acetic acid, ether, aqueous solution of sodium bicarbonate, isobornyl acetate, and sodium sulfate.

Procedure:

0.3012

1. In a long-neck 10 mL round-bottom flask, add a racemic 0.3 g of borneol product with 0.3 mL of glacial acid. Dissolve the racemic mixture in 1 mL of acetone. Add a boiling chip.
2. Assemble a reflux with a distillation column and place the apparatus on a water bath with a temperature between 40-50 C. Make sure the temperature of the water bath does not deviate from the specified range. If the borneol does not dissolve at this temperature, add 0.5 mL of acetone to the mixture until all borneol goes into the solution.
3. Once all reagents are dissolved, add 6 mL of aqueous sodium hypochlorite solution (bleach.) The addition should be done through the top opening of the condenser and should be paced dropwise over 35 minutes. After each addition of the hypochlorite, the mixture should be given a gentle swirl to ensure all chemicals are mixed thoroughly. The reaction must be allowed to continue for 15 minutes after the laster bleach addition.

add slowly

start 1:13pm

End 1:28pm

4. Let the mixture cool to room temperature and extract the camphor with two portions (2mL each) of methylene chloride (or dichloromethane, or DCM.) To make the extraction more efficient, use a centrifuge tube.
5. Neutralize the combined methylene chloride layers with ~2mL of saturated sodium bicarbonate. Keep adding bicarbonate solution if necessary. The presence of acidic residuals will be determined by the production of CO2 bubbles after every addition of the bicarbonate solution.
6. Wash the organic solution sequentially with 2 mL of 5 % sodium bisulfite and with 2 mL of water.
7. Dry the organic solution over anhydrous sodium sulfate. The solution is considered dry when freely rolling crystals of the drying agent are observed.
8. Evaporate the DCM over a water bath with a temp not exceeding 50 C. Weigh the collected crystals and determine the percent yield. Compare the melting point range with that published in literature.

Cleanup:

- ☐ Combine all neutralized aqueous layers and wash down the sink with plenty of water. Use the sinks inside the fume hood. The drying agent can be placed in the waste container or washed down the drain with plenty of water. The residual DCM and all other organic products must be disposed of in the designated waste container.
- ☐ Clean area. Turn off your vacuum line, air line, and the light in your hood before you leave. Make sure that your hood sash is closed. RETURN ALL

GLASSWARE BACK TO YOUR WORKSPACE IN CLEAN AND WORKING ORDER.

# Lab Performance

Exp 6 D3

Hannah Sommers | Chem 2081-003 | 402 | ◯

*Name:* | *Section:* | *Room:* | *Hood:*

| | |
|---|---|
| **Wearing personal protection equipment**—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves. | ○ ○ ○ ○ ○ (5pts) |
| **Safe working habits**—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks. | ○ ○ ○ ○ ○ (5pts) |
| **Following our laboratory safety technique work procedures**—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts). | ○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts) |
| **Glassware operation**—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts). | ○(1pt) ○ (5pts) ○ (5pts) |
| **Working independently** or equal partner participation during the entire lab. | ○ ○ ○ ○ ○ (5pts) |
| **Good work behavior**—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace. | ○ ○ ○ ○ ○ (5pts) |
| **Maintain clean and organized personal space.** | ○ ○ ○ ○ ○ (5pts) |

GTA notes:

*GTA signature:* _____ *Final Performance Grade:* [ ]